

PROJECT DEVELOPMENT PHASE

Leave Approval System Using ServiceNow

1. Development Phase Overview

The Development Phase involves building and configuring the Leave Approval System on the ServiceNow platform. During this phase, all project requirements identified in the analysis and design phases are implemented using ServiceNow components such as Custom Applications, Tables, Forms, Client Scripts, Flow Designer, Approval Workflows, and Email Notifications.

The primary objective of this phase is to develop an automated system that allows employees to submit leave requests and enables managers to approve or reject them efficiently.

2. Development Environment

Component	Description
Platform	ServiceNow Developer Instance
Application Scope	Global
Update Set	Global
Development Tool	Flow Designer
Database	ServiceNow Tables
Notification Service	Email Notifications
Scripting	Client Scripts

3. Module Development

Module 1: Application Development

Description

A custom application named Leave Approval System was created to manage leave requests and approvals.

Activities Performed

- Created a custom application.
- Configured application menu.

- Added modules for navigation.

Output

Centralized application structure for leave management.

Module 2: Database Development

Description

A custom table named Leave Request was created to store leave-related information.

Table Name

Leave Request

Fields Created

Field Name	Type
Employee	Reference(User)
Manager	Reference(User)
Leave Type	Choice
Start Date	Date
End Date	Date
Reason	String
Status	Choice

Manager Comments String

Output

Centralized storage of leave records.

Favorites

History

Workspaces

Admin

Table - Leave Request

Search

Table

Leave Request

Delete

Update

Delete All Records

* Label

Leave Request

* Name

u_leave_request

Application

Global

Remote Table

Columns

Controls

Application Access

Table Columns

Column label

Search

1 to 14 of 14

New

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Created	Date/Time	(empty)	40	false	
Created by	String	(empty)	40	false	
Employee	Reference	User	32	false	
End Date	Date	(empty)	40	false	
Leave Type	Choice	(empty)	40	false	
Manager	Reference	User	32	false	
Manager Comments	String	(empty)	40	false	
Reason	String	(empty)	40	false	
Start Date	Date	(empty)	40	false	
Status	Choice	(empty)	40	false	
Sys ID	Sys ID (GUID)	(empty)	32	false	
Updated	Date/Time	(empty)	40	false	
Updated by	String	(empty)	40	false	
Updates	Integer	(empty)	40	false	

Delete

Update

Delete All Records

Module 3: Form Development

Description

A user-friendly form was designed for employees to submit leave requests.

Form Components

- Employee
- Manager
- Leave Type
- Start Date
- End Date
- Reason
- Status

Output

Easy-to-use interface for submitting leave requests.

Favorites

History

Workspaces

Admin

Leave Request - Create Created

Search

<

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Leave Request

New record

Submit

Employee

Reason

Leave Type

-- None --

Status

-- None --

Start Date

Manager Comments

End Date

Manager

Submit

Module 4: Leave Type Configuration

Description

Choice values were configured for leave categories.

Leave Types

- Sick Leave
- Casual Leave
- Earned Leave
- Emergency Leave
- Work From Home

Output

Standardized leave classification.

Favorites

History

Workspaces

Admin

Leave Request - Create Created

Search

<

≡

Leave Request

New record

Submit

Employee

Reason

Leave Type

-- None --

Status

-- None --

Start Date

Manager Comments

End Date

Manager

Submit

-- None --

Casual Leave

Sick Leave

Work From Home

Earned Leave

Emergency Leave

Module 5: Status Configuration

Description

Status values were configured to track request progress.

Status Values

- Pending
- Approved
- Rejected

Default Value

Pending

Output

Real-time leave request tracking.

The screenshot displays a web application interface for creating a new leave request. The top navigation bar includes links for Favorites, History, Workspaces, and Admin. The main header shows 'Leave Request - Create Created' with a search bar and a 'Submit' button. The form itself is titled 'Leave Request New record' and contains several input fields: Employee (text), Leave Type (dropdown), Start Date (calendar), End Date (calendar), Reason (text), Status (dropdown), Manager Comments (text), and Manager (dropdown). The Status dropdown menu is currently open, showing the following options: -- None --, Pending, Approved, and Rejected. The Manager dropdown menu is also open, showing the same options. A 'Submit' button is located at the bottom left of the form.

Module 6: Date Validation Development

Description

A Client Script was developed to validate leave dates before submission.

Validation Rule

End Date should not be earlier than Start Date

Functionality

- Checks date values.
- Prevents invalid submissions.
- Displays validation message.

Output

Improved data accuracy.

Favorites

History

Workspaces

Admin

Email - Leave Request Approved ☆

Search

Email

Leave Request

Re

Cont

Headers

Update

Delete

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Preview Email

×

Close

Hello TamiTrybus,

Your leave request has been approved.

Leave Type:Sick Leave

Start Date:2026-06-23

End Date:2026-06-24

Ref:MSG0001427_T9z0QQGe47itB6585kmd

Message-ID:<645112886.17.1782197839335@app129173.aus191.service-now.com>

X-ServiceNow-Source:FlowDesigner-310914db6c650794964461731c58d26b

X-ServiceNow-Generated:true

X-ServiceNow-SysEmail-Version:2

Update

Delete